SCHOLARSHIP MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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(Reg.No:24MCR111)

in partial fulfilment of the requirements

for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

IN

DEPARTMENT OF COMPUTER APPLICATIONS

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**SCHOLARSHIP MANAGEMENT SYSTEM**" is the bonafide record of project work done by the student **SUJITHA K C (24MCR111)** in partial fulfilment for the requirements award of the Degree of Master of Computer Applications, Anna university Chennai during the year 2025-2026.

SUPERVISOR **HEAD OF THE DEPARTMENT**

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Date:

Submitted for the end semester viva voce examination held on ____________

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I affirm that the project report entitled **"SCHOLARSHIP MANAGEMENT SYSTEM"** being submitted in partial fulfilment of the requirements for the award of Master of Computer Applications is the original work carried out by me. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Date: **SUJITHA K C**

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I certify that the declaration made by the above candidates is true to the best of my knowledge.

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[Dr.M.SINDHU]

ABSTRACT

The Scholarship Management System is a Salesforce-based application developed to simplify and automate the scholarship management process at an educational institution. Traditionally, scholarship applications and approvals are handled manually through paper forms and spreadsheets, which often leads to inefficiencies such as data redundancy, delayed approvals, difficulty in verifying eligibility, and lack of centralized record management. This project aims to overcome these challenges by providing a centralized digital platform for managing scholarships effectively.

The system allows students to view available scholarship schemes, understand eligibility criteria, and submit applications through an online portal built using Salesforce Experience Cloud and Lightning Web Components (LWC). During the application process, the system collects personal, academic, and financial details from students and verifies eligibility conditions such as family income, academic performance, attendance percentage, community category, and course of study.

On the administrative side, scholarship officers can create and manage scholarship schemes, review student applications, and approve or reject applications based on predefined eligibility rules. Once an application is approved, automated backend processes implemented using Apex triggers and handler classes create or update student records and generate scholarship award records. This automation ensures accurate data management and reduces manual intervention. The platform serves as a reliable and user-friendly solution for both students and administrative staff.

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LIST OF ABBREVATIONS

ABBREVIATION	EXPANSION
SDG	Sustainable Development Goals
DFD	Data Flow Diagram
ER	Entity Relationship
UML	Unified Modeling Language
LWC	Lightning Web Components
UI	User Interface
API	Application Programming Interface
SRS	Software Requirement Specification
DB	Database
SMS	Short Message Service
AI	Artificial Intelligence
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
VS Code	Visual Studio Code

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The Scholarship Management System is a web-based application developed on the Salesforce platform to automate and simplify the scholarship management process at an educational institution. Scholarships play a significant role in supporting students with financial assistance based on their academic performance, economic background, and eligibility criteria. Traditionally, managing scholarships involves manual processes such as collecting applications, verifying eligibility, maintaining records, and approving awards. These manual procedures often result in inefficiencies, delays, and data management issues. In addition, manual handling increases the chances of errors and makes it difficult to track and update records efficiently.

The proposed Scholarship Management System provides a centralized digital platform that simplifies the entire scholarship lifecycle. The system enables students to view available scholarship schemes, verify eligibility criteria, apply for scholarships, and track the status of their applications through an online portal. The portal is developed using Salesforce Experience Cloud and Lightning Web Components (LWC), which provide a responsive and user-friendly interface. It also ensures that students can access the system anytime and from anywhere, improving accessibility and convenience.

On the administrative side, scholarship officers can create and manage scholarship schemes, review student applications, verify eligibility conditions, and approve or reject applications. Automated backend processes implemented using Apex triggers and handler classes ensure that once an application is approved, the system automatically creates or updates student records and generates scholarship award records. This automation improves efficiency, reduces manual workload, and ensures accurate record management. The system enhances transparency, simplifies administrative processes, and provides an efficient platform for managing scholarships within the ins

1.1 EXISTING SYSTEM

In the existing system, scholarship management is primarily handled through manual processes. Students typically obtain scholarship application forms from the college administration office, fill them manually, and submit the forms along with supporting documents such as income certificates, academic records, and community certificates. The submitted applications are then manually reviewed by administrative staff.

The verification process involves checking eligibility criteria such as family income, academic performance, attendance percentage, and community category. Since these processes are performed manually, it often leads to delays and errors in application processing. Data related to scholarship applications is usually maintained in spreadsheets or paper records, which makes it difficult to track and manage information efficiently.

Additionally, manual record maintenance makes it challenging to maintain historical scholarship data and generate reports when required. The lack of a centralized digital platform results in inefficiencies and increases the administrative workload for scholarship officers.

1.2 DRAWBACKS OF EXISTING SYSTEM

The existing manual scholarship management system has several limitations that affect efficiency and data management. The major drawbacks are listed below:

- Scholarship applications are processed manually using paper forms.
- Maintaining records in spreadsheets or physical files leads to data redundancy.
- Verification of eligibility criteria takes a long time.
- There is a high chance of human errors during data entry and verification.
- Students cannot easily track the status of their scholarship applications.
- Lack of centralized data management makes it difficult to maintain scholarship records.
- Generating reports and analyzing scholarship data is difficult.
- Manual processes increase the workload of administrative staff.

1.3 PROPOSED SYSTEM

To overcome the limitations of the existing system, a Salesforce-based Scholarship Management System is proposed. The system provides an automated and centralized platform for managing scholarship schemes, student applications, and scholarship awards.

The proposed system enables students to access a web-based portal built using Salesforce Experience Cloud. Through this portal, students can browse available scholarship schemes, review eligibility criteria, and submit applications online. The application form collects essential information such as personal details, academic records, financial details, and other required information.

The system also includes an eligibility checking mechanism implemented using Lightning Web Components and Apex. This feature allows the system to automatically evaluate student eligibility based on predefined criteria such as family income, minimum marks, attendance percentage, gender eligibility, community eligibility, and study year.

On the administrative side, scholarship officers can manage scholarship schemes and review student applications through the Salesforce platform. Once an application is approved, automated backend processes using Apex triggers and handler classes create or update student records and generate scholarship award records. This automation ensures data consistency and reduces manual intervention.

The system architecture consists of an Experience Cloud portal as the user interface, Lightning Web Components for frontend functionality, Apex controllers for business logic, and Salesforce custom objects for data storage. This architecture ensures scalability, reliability, and efficient data management. The system also improves transparency in scholarship allocation and enables administrators to easily track scholarship records. Furthermore, it provides a user-friendly interface that simplifies the application process for students and reduces administrative workload. Overall, the proposed system enhances efficiency and ensures a structured approach to scholarship management within the institution.

1.4 ADVANTAGES OF PROPOSED SYSTEM

The proposed Scholarship Management System provides several advantages by automating the scholarship management process. The key benefits of the system include:

- Provides a centralized platform for managing scholarships.
- Allows students to view available scholarships and check eligibility easily.
- Enables students to apply for scholarships through an online portal.
- Automates eligibility verification based on predefined criteria.
- Reduces manual workload for administrative staff.
- Prevents duplicate student records through automated data management.
- Improves transparency in the scholarship allocation process.
- Provides faster processing and approval of scholarship applications.
- Maintains accurate and organized scholarship records.

SUMMARY

The Scholarship Management System is designed to automate the scholarship management process at Kongu Engineering College using the Salesforce platform. The system replaces the traditional manual process with a centralized digital solution that enables students to browse scholarships, verify eligibility, apply online, and track application status. Administrative users can manage scholarship schemes, review applications, and approve awards efficiently. By integrating Salesforce Experience Cloud, Lightning Web Components, Apex controllers, and automated triggers, the system ensures efficient data management and streamlined scholarship processing. The proposed system enhances transparency, reduces administrative workload, and provides an efficient solution for managing scholarships within the institution.

CHAPTER 2

SYSTEM ANALYSIS

2.1 IDENTIFICATION ON NEED

In educational institutions, scholarships play a vital role in supporting students who require financial assistance. However, the traditional process of managing scholarship applications often involves manual paperwork, spreadsheets, and multiple verification steps handled by administrative staff. These manual processes are time-consuming, error-prone, and inefficient, leading to delays in application processing and difficulties in maintaining accurate records.

At the institution, managing scholarship schemes and student applications through manual methods creates challenges such as data redundancy, difficulty in verifying eligibility conditions, and lack of centralized data storage. Students also face difficulties in accessing information about available scholarships and tracking the status of their applications.

Therefore, there is a need for an automated system that can manage scholarship schemes, student applications, eligibility verification, and scholarship awards in an efficient and organized manner. The proposed Scholarship Management System addresses these needs by providing a centralized web-based platform developed using Salesforce. The system simplifies scholarship management by enabling students to apply online and allowing administrators to manage scholarship processes through automated workflows.

2.2 FEASIBILITY STUDY

A feasibility study is conducted to determine whether the proposed system is practical, cost-effective, and technically achievable. It helps in evaluating the overall viability of the system before implementation and ensures that the proposed solution meets the requirements of the organization. Through feasibility analysis, the benefits, limitations, and possible risks of implementing the system are carefully examined.

The feasibility study for the Scholarship Management System focuses on analyzing whether the system can be successfully developed and implemented within the institution. It ensures that the system will operate efficiently while meeting the needs of both students and administrative staff.

The feasibility analysis mainly includes the following types:

- Economic feasibility
- Operational feasibility
- Technical feasibility

These aspects evaluate the financial viability, usability, and technological capability of the system. In addition to these, schedule feasibility examines whether the system can be developed within the available time frame, while legal feasibility ensures compliance with institutional policies and data security regulations.

2.2.1 Economic Feasibility

Economic feasibility evaluates whether the proposed system is financially viable and whether the benefits of the system outweigh its implementation costs. Since the Scholarship Management System is developed on the Salesforce platform, the primary costs involve platform access and system development efforts.

The implementation of this system significantly reduces the operational costs associated with manual processing of scholarship applications, maintaining paper records, and performing repetitive administrative tasks. Automation of eligibility verification and scholarship approval reduces the time and effort required by administrative staff. In the long term, the system provides cost savings by improving efficiency and minimizing manual errors.

2.2.2 Operational Feasibility

Operational feasibility examines how well the proposed system will function within the existing organizational environment. The Scholarship Management System is designed to be user-friendly and accessible for both students and administrators.

Students can easily browse available scholarship schemes, check eligibility criteria, and submit applications through the Experience Cloud portal. Administrative users can review applications, verify eligibility, and manage scholarship schemes through the Salesforce platform. The system simplifies the workflow of scholarship management and reduces manual effort. Since the system provides a clear interface and automated processes, it is expected to be easily adopted by users within the institution.

2.2.3 Technical Feasibility

Technical feasibility evaluates whether the required technology and resources are available to implement the system successfully. The Scholarship Management System is developed using Salesforce technologies such as Lightning Web Components (LWC), Apex programming, Experience Cloud, and Salesforce custom objects.

These technologies provide a scalable and reliable platform for developing secure web applications. Salesforce also offers built-in features such as security, automation, and data management, which support efficient system implementation.Since the required technologies are readily available and widely used in modern enterprise applications, the system is technically feasible to implement and maintain.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specification (SRS) defines the hardware and software requirements necessary for implementing the system. It describes the technical environment required for developing, deploying, and operating the Scholarship Management System.

2.3.1 Hardware Requirements

The hardware requirements for the system are minimal since the application is developed and hosted on the Salesforce cloud platform. The following hardware configuration is sufficient for accessing the system:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 20 GB free disk space
- Display: Standard monitor with 1024 × 768 resolution
- Internet Connection: Stable broadband internet connection

2.3.2 Software Requirements

The software requirements necessary for developing and accessing the system include:

- Operating System: Windows 10 or higher / macOS / Linux
- Platform: Salesforce Platform
- Frontend Technology: Lightning Web Components (LWC)
- Backend Technology: Apex Programming Language
- Development Tools: Salesforce Developer Console / Visual Studio Code with Salesforce Extensions
- Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge
- Portal Framework: Salesforce Experience Cloud

2.4 SOFTWARE DESCRIPTION

The Scholarship Management System is built using a layered architecture consisting of frontend components, backend logic, and a centralized database. The system integrates Salesforce Experience Cloud, Lightning Web Components, and Apex programming to provide a complete solution for scholarship management. It ensures smooth communication between user interfaces and server-side processes, enabling efficient data handling and interaction between different system modules.

The architecture supports scalability, security, and maintainability, allowing the system to handle multiple users simultaneously while ensuring reliable performance and consistent data processing. It also improves system flexibility, making it easier to update features and adapt to future requirements. This structured design helps in efficient system management and enhances overall performance and user experience.

Additionally, the system promotes better data organization and reduces redundancy across different modules. It ensures faster processing of applications and accurate record maintenance. The design also supports easy integration of new features and future enhancements. Overall, it provides a reliable and efficient platform for managing scholarships effectively.

2.4.1 Front End

The frontend of the Scholarship Management System is developed using Lightning Web Components (LWC) within the Salesforce Experience Cloud environment. The frontend serves as the user interface through which students interact with the system and perform scholarship-related activities. It provides an intuitive and responsive portal that allows students to easily access scholarship information and submit applications.

The frontend includes various components such as scholarship listing, scholarship details, eligibility recommendation, and scholarship application forms. These components enable students to browse available scholarships, review eligibility criteria, and apply for suitable scholarships through a structured form. The user interface is designed to provide clear navigation and smooth interaction for users.

The portal also includes customized elements such as banners, logos, and navigation menus to enhance the user experience. Static resources are used for branding purposes, and the portal layout is managed using Salesforce Experience Builder. Overall, the frontend provides a visually appealing and user-friendly interface for accessing scholarship services.

2.4.2 Back End

The backend of the Scholarship Management System is implemented using Apex programming in the Salesforce platform. Apex controllers manage the business logic required for processing scholarship applications and verifying eligibility conditions. The backend ensures that all application data is processed correctly and stored securely within the system.

Automation plays a key role in the backend architecture. Apex triggers and handler classes are used to automate processes such as verifying student records, creating scholarship awards, and updating application status. When a scholarship application is approved, the trigger automatically creates or updates the corresponding student record and generates a scholarship award record.

The backend architecture is designed to be scalable and maintainable. The use of handler classes ensures clean code structure and prevents duplication of logic. Salesforce security features also help protect sensitive student information and ensure controlled access to data.

2.4.3 Database

The database for the Scholarship Management System is managed using Salesforce custom objects, which store all scholarship-related information in a structured format. The database maintains records for scholarship schemes, student applications, student details, and scholarship awards.

Key objects in the system include Scholarship_Scheme__c, Scholarship_Application__c, Student__c, and Scholarship_Award__c. These objects are interconnected through relationships that allow the system to track scholarship applications and awards for each student. This structured data model ensures efficient data storage and retrieval.

Salesforce provides secure cloud-based data management, ensuring data integrity and reliability. The platform supports real-time updates, which allows administrators to easily track scholarship records and generate reports when required. The database design ensures that scholarship information is organized, accessible, and maintained efficiently.

SUMMARY

The system analysis phase evaluates the requirements and feasibility of implementing the Scholarship Management System. The analysis identifies the need for automating the scholarship management process and examines the economic, operational, and technical feasibility of the proposed solution. The chapter also defines the hardware and software requirements necessary for system implementation and provides an overview of the system architecture, including frontend, backend, and database components. The proposed system ensures efficient scholarship management through automation, centralized data storage, and a user-friendly web-based platform.

CHAPTER 3

SYSTEM DESIGN

3.1 MODULE DESCRIPTION

The Scholarship Management System is designed using a modular architecture to ensure better organization, maintainability, and scalability of the system. Each module performs a specific function in the scholarship management process, allowing the system to efficiently handle scholarship schemes, student applications, eligibility verification, and scholarship awards. The modular design also simplifies development, testing, and future enhancements of the system.

3.1.1 Scholarship Scheme Module

The Scholarship Scheme Module is responsible for managing all scholarship schemes available in the system. This module allows administrators to create, update, and manage scholarship schemes offered by the institution. Each scholarship scheme contains important eligibility details such as course eligibility, admission type, eligible study year, family income limit, minimum academic marks, attendance requirements, gender eligibility, and community eligibility. This module stores scholarship scheme information in the Scholarship_Scheme__c object and ensures that only active scholarships are displayed to students through the portal.

3.1.2 Scholarship Application Module

The Scholarship Application Module allows students to apply for scholarships through the online portal. Students can submit their personal, academic, and financial details through an application form developed using Lightning Web Components.The application form collects important information such as student name, roll number, course details, family income, attendance percentage, and community category. All application data is stored in the Scholarship_Application__c object. This module ensures that applications are submitted and recorded properly for further verification and approval.

3.1.3 Eligibility Evaluation Module

The Eligibility Evaluation Module is responsible for determining whether a student qualifies for a particular scholarship scheme. The module evaluates student information against predefined eligibility criteria defined in the scholarship scheme.Eligibility conditions such as family income, academic performance, attendance percentage, gender eligibility, community eligibility, and study year are automatically verified using Apex logic. Based on this evaluation, the system determines whether the student is eligible or not eligible for a specific scholarship. This automated verification process reduces manual effort and improves accuracy.

3.1.4 Student Module

The Student Module manages student records within the system. When a scholarship application is approved, the system either creates a new student record or updates an existing student record based on the roll number.Student details such as name, roll number, course, academic year, gender, community, income details, and contact information are stored in the Student__c object. This module ensures that student data is maintained accurately and prevents duplicate records in the system.

3.1.5 Approved Scholarship Module

The Approved Scholarship Module handles the management of scholarships that have been approved for students. Once an application is approved, the system automatically creates a scholarship award record.This module stores details such as the student receiving the scholarship, the scholarship scheme, the academic year, the approved date, and the scholarship amount. The data is stored in the Scholarship_Award__c object. This module helps administrators track scholarship distributions and maintain historical scholarship records.

3.1.6 Notification Module

The Notification Module is responsible for informing students and administrators about important updates in the scholarship process. Notifications may include application submission confirmation, eligibility results, application approval or rejection, and scholarship award information.The notification mechanism helps keep users informed about the status of scholarship applications and improves communication between students and administrators. This module enhances transparency and ensures that users receive timely updates regarding scholarship processing.

3.2 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) represents the flow of data within the system and illustrates how information moves between different components. It provides a graphical representation of processes, data stores, and external entities involved in the system, helping to understand how data is processed, stored, and transferred within the Scholarship Management System. The DFD for the proposed system shows how students interact with the scholarship portal, submit applications, and how the system processes the data through eligibility evaluation and approval mechanisms, while also illustrating the interaction between students, administrators, and the system database.

3.2.1 DFD Level 0

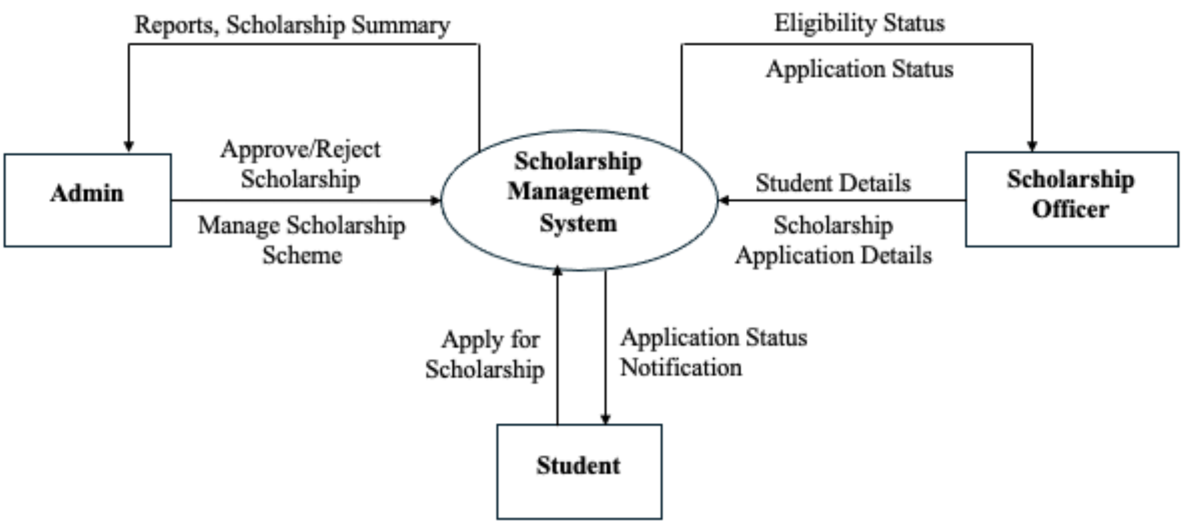

Figure 3.1 Data Flow Diagram Level 0

The Level 0 Data Flow Diagram (DFD), also known as the context diagram, provides a high-level overview of the Scholarship Management System. It represents the entire system as a single process and shows how the system interacts with external entities.

In the Figure 3.1 Data Flow Diagram, the main external entities interacting with the system are Student, Admin, and Scholarship Officer. Students interact with the system by applying for scholarships and receiving notifications about the status of their applications. The Admin manages scholarship schemes and has the authority to approve or reject scholarship applications. The Scholarship Officer reviews student details, verifies application information, and monitors the eligibility status of students.

The system processes the information provided by these entities and generates outputs such as scholarship summaries, eligibility results, application status updates, and reports. This diagram provides a clear overview of how data flows between users and the system at the highest level.

3.2.2 DFD Level 1

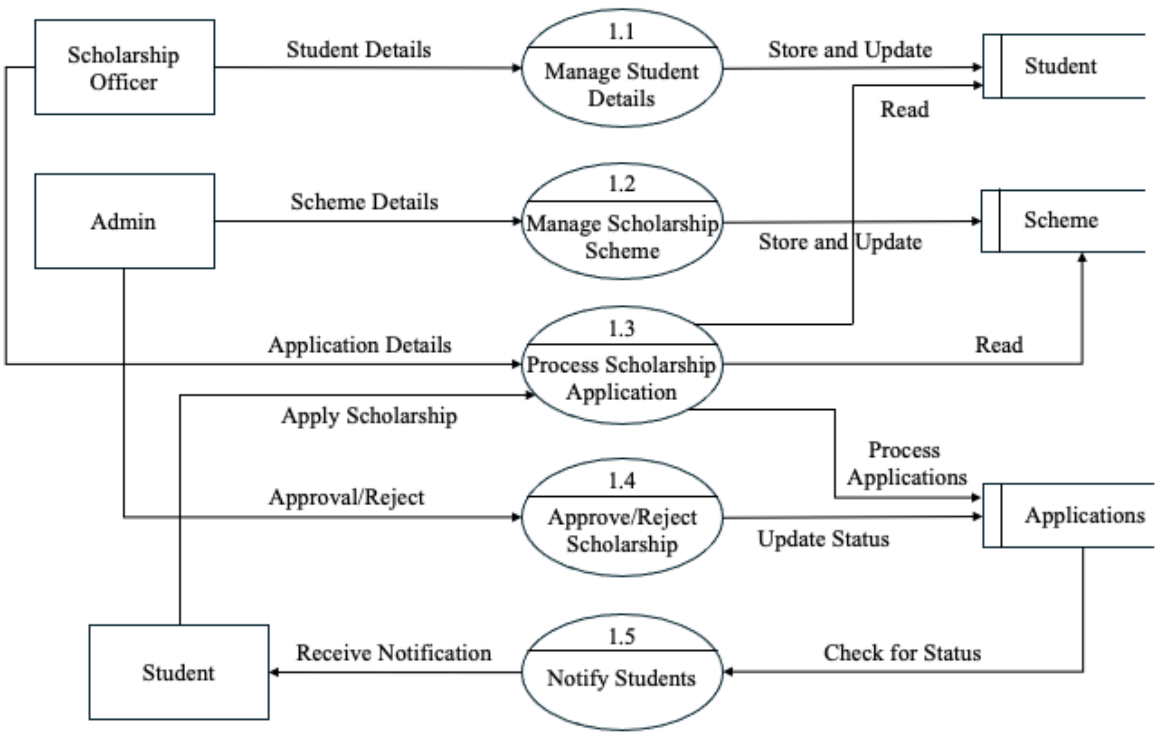

Figure 3.2 Data Flow Diagram Level 1

The Level 1 Data Flow Diagram provides a detailed representation of the internal processes of the Scholarship Management System. It breaks down the main system process into smaller functional modules that handle specific operations such as managing student data, scholarship schemes, application processing, approval, and notification. The diagram also illustrates how data flows between external entities, processes, and data stores.

1.1 Manage Student Details

- Handles the management of student information in the system.
- The Scholarship Officer enters student details such as name, roll number, course, and academic year.
- The system processes the information provided.
- Student records are stored or updated in the Student data store.
- Ensures that student data is maintained accurately for scholarship processing.

1.2 Manage Scholarship Scheme

- Allows the Admin to create and manage scholarship schemes.
- Admin enters scheme details such as eligibility criteria, income limit, course eligibility, and scholarship amount.
- The system processes the scheme information.
- The details are stored and updated in the Scheme data store.
- Helps maintain structured information about available scholarship schemes.

1.3 Process Scholarship Application

- Students submit scholarship applications through the portal.
- The system collects application details from the students.
- Information includes student details, selected scholarship scheme, and supporting information.
- The application data is processed by the system.
- Application records are stored in the Applications data store for verification.

1.4 Approve/Reject Scholarship

- Admin reviews the submitted scholarship applications.
- Application details are verified based on eligibility criteria.
- Admin decides whether to approve or reject the application.
- The system updates the decision in the Applications data store.
- This step finalizes the application processing stage.

1.5 Notify Students

- The system checks the updated application status.
- Notification is generated based on approval or rejection.
- Students receive information about their application result.
- The notification ensures transparency in scholarship processing.
- Students can view the updated application status through the portal.

3.3 SYSTEM FLOW DIAGRAM

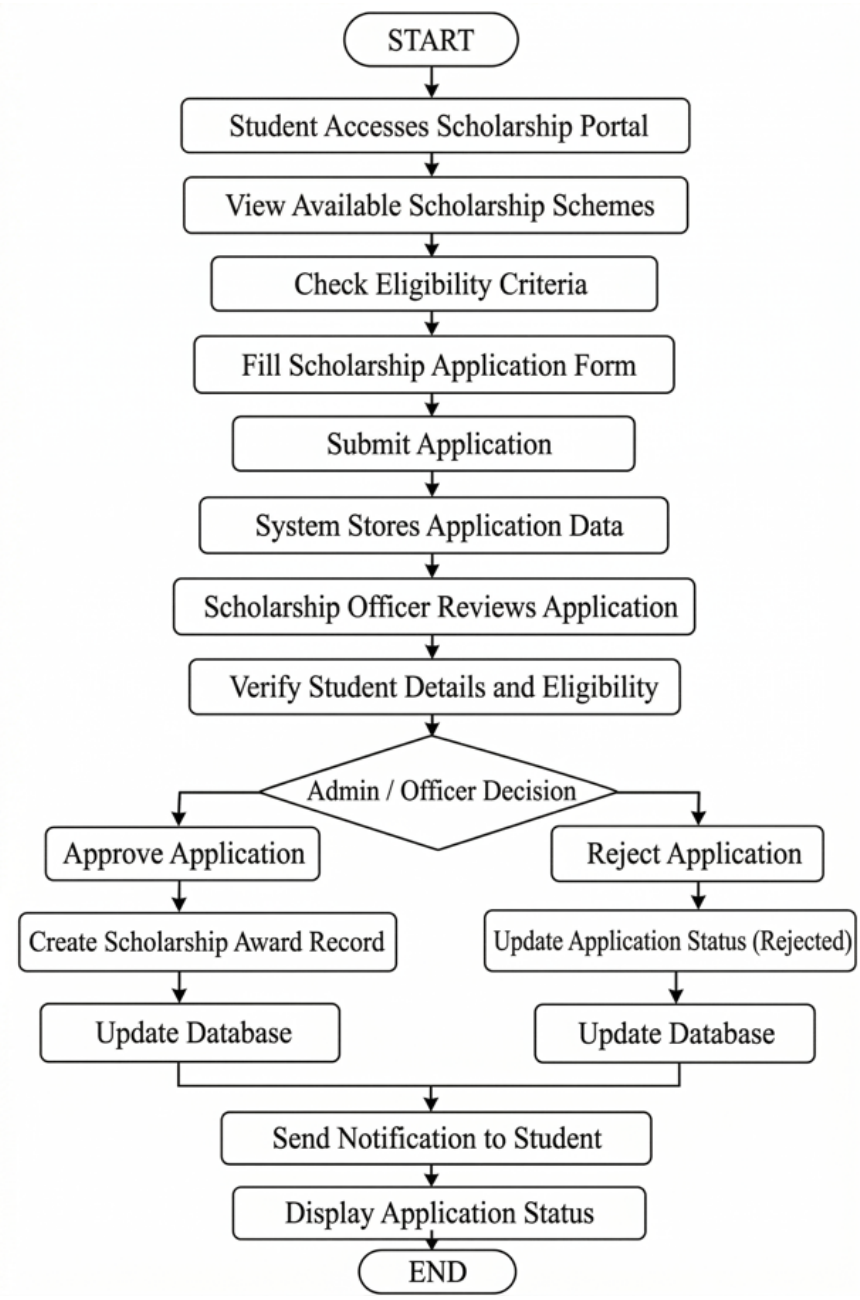

Figure 3.3 System Flow Diagram of Scholarship Management System

The System Flow Diagram illustrates the sequence of operations performed in the Scholarship Management System. The process begins when a student accesses the scholarship portal and views available scholarship schemes. The student can check eligibility criteria and fill the scholarship application form. After submission, the system stores the application data in the database.

The scholarship officer reviews the submitted application and verifies the student details and eligibility conditions. Based on the verification, the admin or scholarship officer makes a decision to approve or reject the application. If approved, the system creates a scholarship award record and updates the database. If rejected, the application status is updated accordingly. Finally, the system sends a notification to the student and displays the application status.

3.4 INPUT DESIGN

Input design focuses on collecting accurate and valid data from users so that the system can process scholarship applications effectively. The Scholarship Management System uses structured input forms developed using Lightning Web Components to collect information from students and administrative users. Input validation mechanisms ensure that only valid data is accepted by the system.

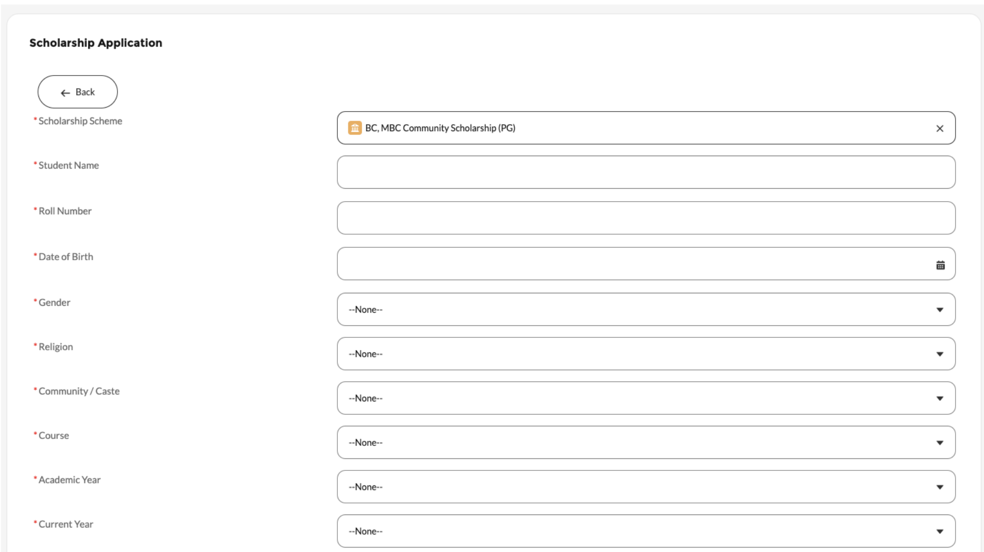

Figure 3.4 Scholarship Application Form

This screen allows students to enter personal, academic, and financial information required to apply for scholarships. The form includes fields such as student name, roll number, course, family income, attendance percentage, community category, and scholarship scheme selection. Proper validations ensure that the entered information follows the required format.

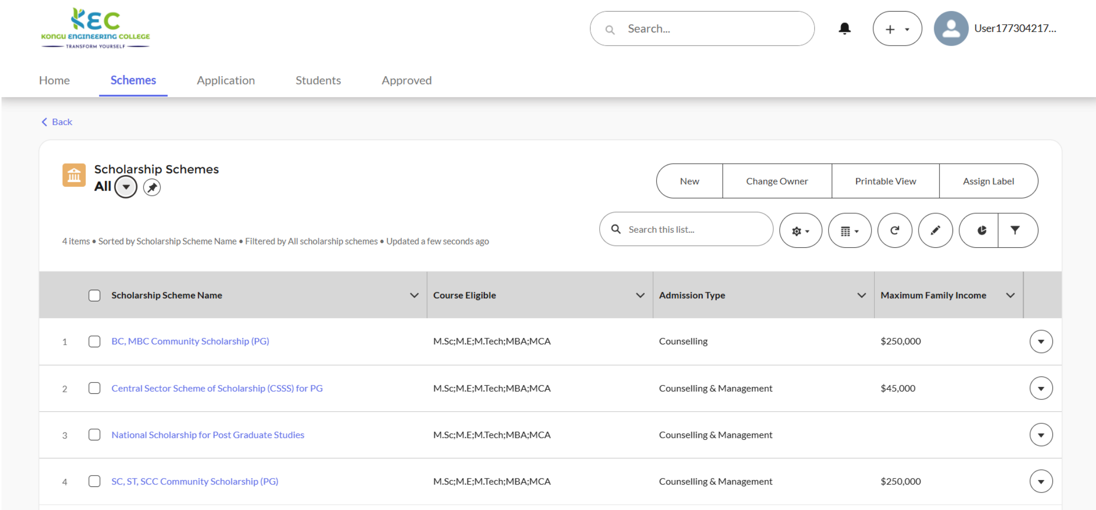

Figure 3.5 Scholarship Scheme Management Screen

This screen is used by the admin or scholarship officer to create and manage scholarship schemes. It includes input fields such as scholarship name, eligibility criteria, course eligibility, minimum marks requirement, maximum family income, and scholarship amount.

Figure 3.6 Scholarship Officer Login Screen

This screen allows authorized users to securely access the system by providing their login credentials. It includes fields such as username and password for authentication. This ensures secure access and prevents unauthorized entry into the system. It also helps maintain data privacy and protects sensitive user information.

3.5 OUTPUT DESIGN

Output design focuses on presenting processed information in a clear and meaningful format for users. The system generates outputs that help students view scholarship details and track application status. Administrative users can view student applications, verify eligibility, and manage scholarship records through output screens. It ensures that information is displayed in an organized and user-friendly manner for easy understanding. Additionally, it supports effective decision-making by providing accurate and timely data.

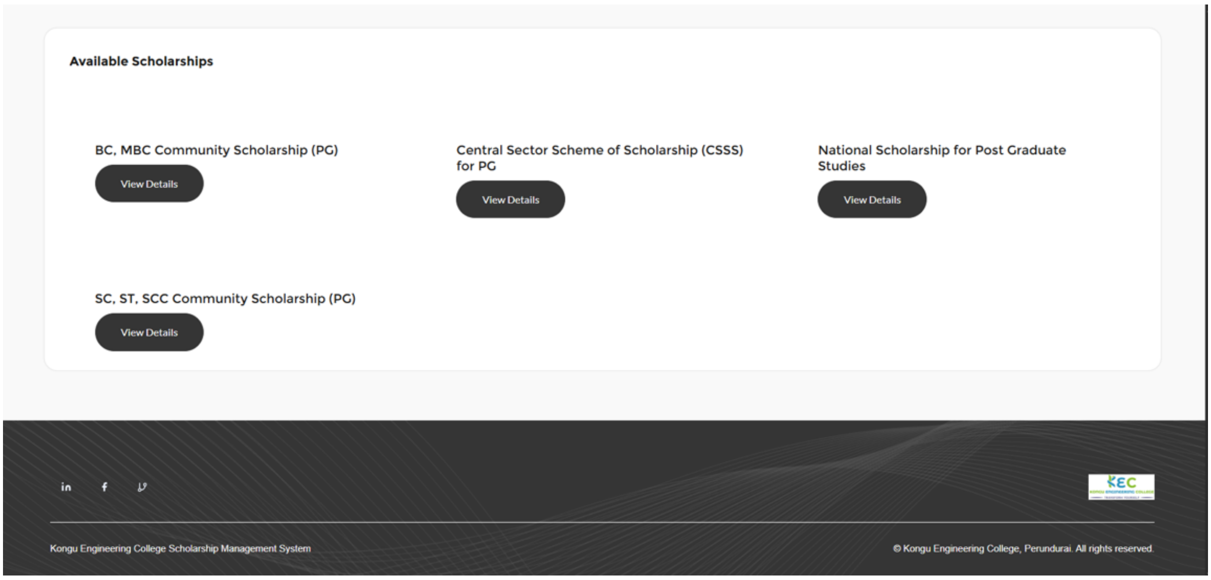

Figure 3.7 Scholarship List Page

This page displays all available scholarship schemes with details such as scholarship name, eligibility criteria, and description. Students can browse available scholarships through this screen. It helps students easily compare different schemes and select the most suitable one based on their qualifications.

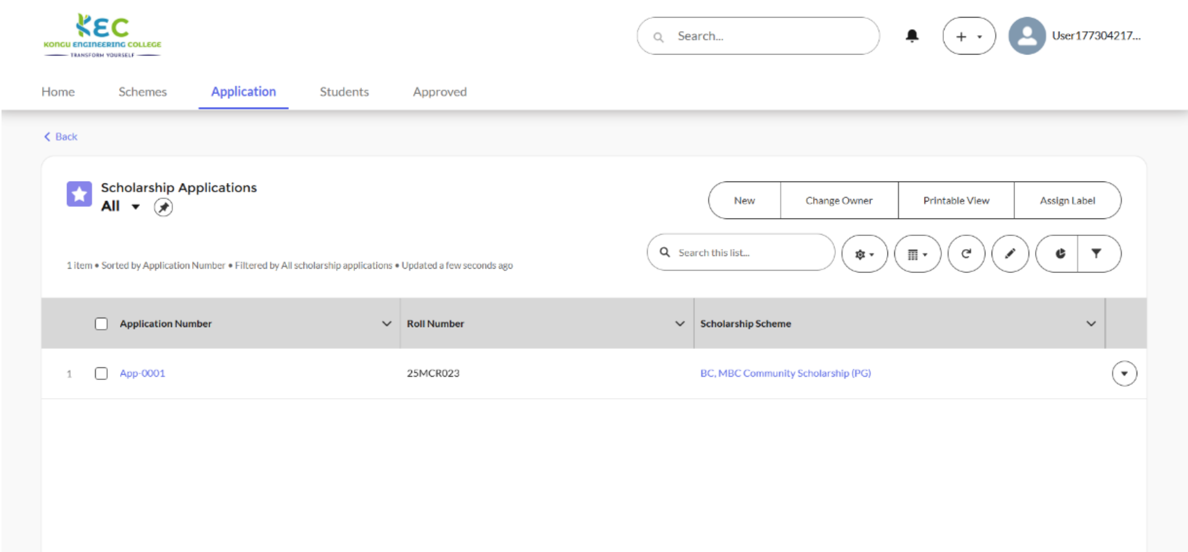

Figure 3.8 Student Application Details Page

This page shows the submitted scholarship application details including student information and selected scholarship scheme. It provides a clear overview of the application for verification and review purposes.

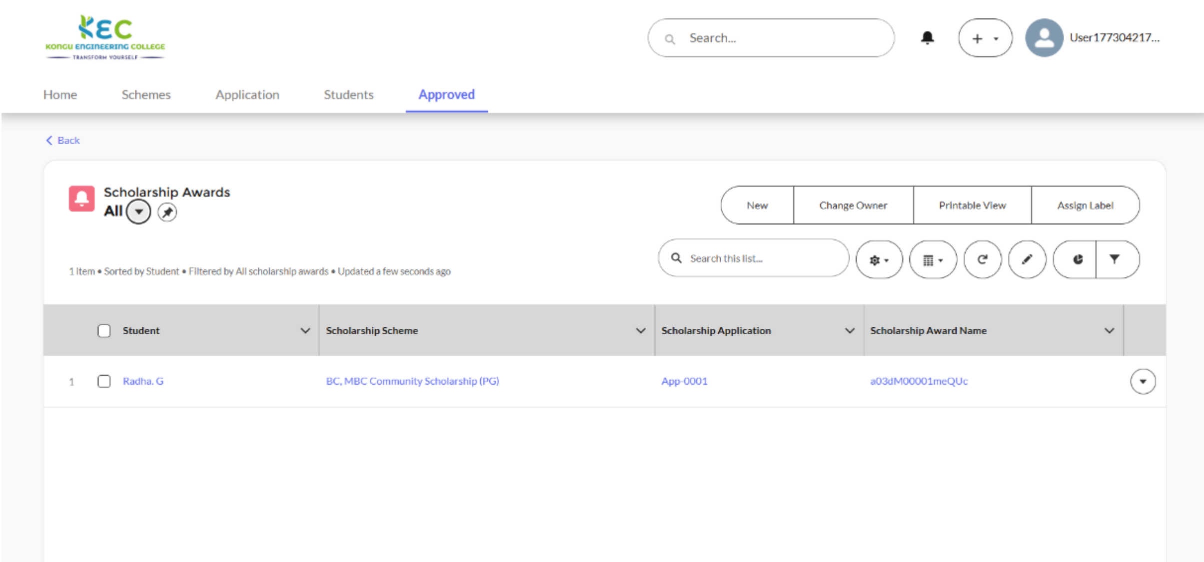

Figure 3.9 Scholarship Award Page

This page displays the scholarships awarded to students after the approval process is completed. It includes important details such as the scholarship amount and approval date. The page helps users easily track and review awarded scholarships.

3.6 ER DIAGRAM

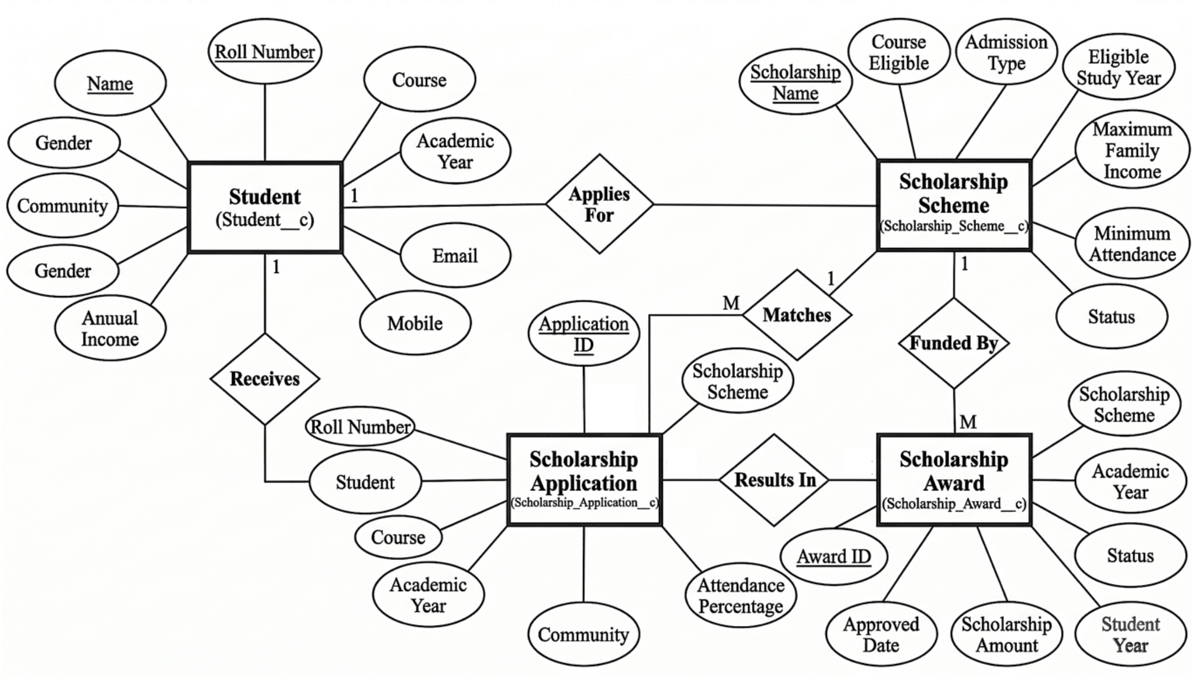

Figure 3.10 Entity Relationship Diagram

The Entity Relationship (ER) Diagram represents the logical structure of the Scholarship Management System by illustrating how different data components are interconnected within the system. It highlights the primary entities such as Student, Scholarship Application, Scholarship Scheme, and Scholarship Award, along with their attributes and relationships. By organizing data into well-defined entities and connections, the diagram provides a clear visualization of how information is stored, accessed, and processed. This structured representation reduces complexity and helps in understanding the overall workflow of the system, making it easier to design, implement, and maintain a scalable database.

The Student entity contains essential details about each applicant, including roll number, name, gender, community, annual income, course, academic year, email, and mobile number. The Scholarship Application entity records the details of applications submitted by students, such as application ID, course, academic year, community, and attendance percentage. Students apply for scholarships through this entity, establishing a relationship between Student and Scholarship Application. The Scholarship Scheme entity defines various scholarship programs with attributes like scholarship name, course eligibility, admission type, eligible study year, income limits, attendance requirements, and status. Applications are matched with suitable schemes based on eligibility criteria, ensuring accurate processing and validation.

The Scholarship Award entity represents the final outcome of the application process, including details such as award ID, approved date, scholarship amount, academic year, student year, and status. It is linked to both the Scholarship Application and Scholarship Scheme entities, indicating that awards are granted based on approved applications and funded schemes. Relationships such as “Applies For,” “Receives,” “Matches,” “Results In,” and “Funded By” define how entities interact, typically following one-to-many associations. Overall, the ER diagram ensures data consistency, reduces redundancy, and supports efficient data retrieval, serving as a strong foundation for building a reliable and well-structured Scholarship Management System.

3.7 DATABASE DESIGN

The database design of the Scholarship Management System is implemented using Salesforce custom objects. These objects store structured data related to scholarship schemes, student applications, student information, and scholarship awards. The database design ensures efficient storage, retrieval, and management of scholarship-related information.

The main objects used in the system include Scholarship_Scheme__c, Scholarship_Application__c, Student__c, and Scholarship_Award__c. Relationships between these objects ensure that scholarship applications are linked to students and scholarship schemes properly. The database design improves data integrity and helps administrators manage scholarship information efficiently.

3.8 TABLE DESIGN

The table design defines the structure of database tables used in the system, where each table contains fields that store specific information related to scholarships and students. It helps in organizing data efficiently for easy storage and retrieval. A well-designed table structure ensures data consistency and minimizes redundancy. It also improves the performance of database operations. Overall, it supports the smooth functioning of the Scholarship Management System.

Table 3.1 Student Object (Student__c)

Field Name	API Name	Data Type
Name	Name	Text
Roll Number	Roll_Number__c	Text
Course	Course__c	Text
Academic Year	Academic_Year__c	Text
Community	Community__c	Picklist
Gender	Gender__c	Picklist
Annual Income	Annual_Income__c	Currency
Email	Email__c	Email
Mobile	Mobile__c	Phone

The Student object stores the details of students who apply for scholarships through the system. This table contains important information such as the student's name, roll number, course, academic year, community category, gender, annual income, email address, and mobile number. It helps in maintaining a complete record of each student for verification and processing. The stored data is used to evaluate eligibility and ensure accurate scholarship allocation.

Table 3.2 Scholarship Scheme Object (Scholarship_Scheme__c)

Field Name	API Name	Data Type
Scholarship Name	Name	Text
Course Eligible	Course_Eligible__c	Picklist
Admission Type	Admission_Type__c	Picklist
Eligible Study Year	Eligible_Study_Year__c	Picklist
Maximum Family Income	Maximum_Income__c	Currency
Minimum Marks Required	Minimum_Marks__c	Number
Minimum Attendance	Minimum_Attendance__c	Percent
Status	Status__c	Picklist

The Scholarship Scheme object stores information related to scholarship programs available in the system. This table includes details such as scholarship name, eligible course, admission type, eligible study year, maximum family income limit, minimum marks required, minimum attendance requirement, and the status of the scholarship scheme. These fields define the eligibility criteria that students must satisfy in order to apply for a particular scholarship. The scholarship scheme table helps administrators manage and maintain all scholarship schemes offered within the institution. It also ensures that eligibility rules are clearly defined and consistently applied across all applications.

Table 3.3 Scholarship Application Object (Scholarship_Application__c)

Field Name	**API Name**	**Data Type**
Application ID	Name	Auto Number
Student Name	Student_Name__c	Lookup(Student)
Roll Number	Roll_Number__c	Text
Course	Course__c	Text
Academic Year	Academic_Year__c	Text
Family Income	Family_Income__c	Currency
Attendance Percentage	Attendance__c	Percent
Community	Community__c	Picklist
Scholarship Scheme	Scholarship_Scheme__c	Lookup

The Scholarship Application object stores the application details submitted by students. This table records information such as application ID, student name, roll number, course, academic year, family income, attendance percentage, community category, and the selected scholarship scheme. The information stored in this table is used by administrators to review applications and verify eligibility conditions. This object also helps track the status of each scholarship application throughout the processing workflow. It ensures that all application data is properly organized and easily accessible. It also supports efficient decision-making during the approval process. Additionally, it helps maintain accurate records for future reference and reporting.

Table 3.4 Scholarship Award Object (Scholarship_Award__c)

Field Name	API Name	Data Type
Award ID	Name	Auto Number
Student	Student__c	Lookup(Student)
Scholarship Scheme	Scholarship_Scheme__c	Lookup
Scholarship Application	Scholarship_Application__c	Lookup
Academic Year	Academic_Year__c	Text
Approved Date	Approved_Date__c	Date
Scholarship Amount	Scholarship_Amount__c	Currency
Status	Status__c	Picklist

The Scholarship Award object maintains records of scholarships that have been approved and granted to students. This table includes details such as award ID, student reference, scholarship scheme, scholarship application reference, academic year, approval date, scholarship amount, and award status. These records help administrators monitor scholarship distribution and maintain historical data of scholarships awarded to students. The scholarship award table ensures proper tracking and management of scholarship benefits provided to eligible students.

SUMMARY

The Scholarship Award object maintains records of scholarships that have been approved and granted to students. This table includes details such as award ID, student reference, scholarship scheme, scholarship application reference, academic year, approval date, scholarship amount, and award status. These records help administrators monitor scholarship distribution and maintain historical data of scholarships awarded to students. The scholarship award table ensures proper tracking and management of scholarship benefits provided to eligible students.

CHAPTER 4

IMPLEMENTATION

4.1 CODE DESCRIPTION

The Scholarship Management System is developed using the Salesforce platform by integrating Lightning Web Components (LWC), Apex programming, and Salesforce custom objects. System architecture follows a modular design where each module performs a specific function related to scholarship management. Implementation ensures efficient handling of scholarship schemes, student applications, eligibility verification, and scholarship awards.

Frontend is developed using Lightning Web Components, providing an interactive interface for students to browse scholarship schemes, check eligibility, and submit applications through the Experience Cloud portal. Components such as scholarshipList, scholarshipDetail, scholarshipApplicationForm, and scholarshipRecommendation handle portal interactions.

Backend logic is implemented using Apex classes and triggers, where controllers process data and perform validation and application processing. Apex Trigger named ScholarshipApplicationTrigger executes when application status is updated. It ensures automation of key processes within the system. It helps maintain consistency and accuracy in application handling.

Trigger calls the ScholarshipApplicationHandler class to process approvals, updating existing student records or creating new ones if needed. After processing, a scholarship award record is generated and stored in the database. System uses Salesforce custom objects such as Scholarship_Scheme__c, Scholarship_Application__c, Student__c, and Scholarship_Award__c to manage schemes, applications, student details, and awards. This structured approach ensures efficient data storage and retrieval.

4.2 STANDARDIZATION OF THE CODEING

Code standardization ensures that the developed system follows consistent coding practices, making the code easier to understand, maintain, and extend. In the Scholarship Management System, coding standards are followed while developing Lightning Web Components and Apex classes. Meaningful variable names, proper indentation, and modular coding practices are used to maintain code readability and quality.

The code is structured using controller classes, handler classes, and triggers to separate business logic from database operations. This approach improves maintainability and allows the system to handle large volumes of data efficiently.

```
public class ScholarshipService {
   public static Boolean checkEligibility(Decimal marks, Decimal income){
      if(marks >= 75 && income <= 200000){
         return true;
      }else{
         return false;
      }
   }
}
```

This code checks whether a student satisfies the eligibility criteria based on marks and family income.

4.3 ERROR HANDLING

Error handling is an important part of system development that ensures the system can handle unexpected situations without crashing. Proper error handling improves system reliability and provides meaningful feedback to users when an error occurs.

In the Scholarship Management System, exception handling techniques are used in Apex code to detect and manage runtime errors. The system uses try-catch blocks to handle exceptions and ensure that errors are logged and appropriate messages are displayed to the user.

```
public static void createStudentRecord(Student__c studentRec){
   try{
      insert studentRec;
      System.debug('Student record created successfully');
   }
   catch(DmlException e){
      System.debug('Error while creating student record: ' + e.getMessage());
   }
}
```

This code demonstrates error handling using a try–catch block in Apex. The system attempts to insert a student record into the database. If an error occurs during the database operation, the exception is caught and an error message is displayed using the debug statement. This approach prevents the system from crashing and helps developers identify and resolve errors effectively.

SUMMARY

This chapter presented the implementation details of the Scholarship Management System developed using the Salesforce platform. The system integrates Lightning Web Components for the frontend interface and Apex programming for backend processing to manage scholarship schemes, student applications, and scholarship awards efficiently. Standardized coding practices are followed to ensure code readability, maintainability, and scalability. Error handling mechanisms are also implemented to manage exceptions and improve system reliability. Overall, the implementation provides an automated and structured solution for handling scholarship processes effectively within the institution.

CHAPTER 5

TESTING AND RESULTS

5.1 TESTING

Testing is an important phase in the system development life cycle that ensures the system functions correctly and meets the specified requirements. It involves evaluating the system to identify errors, verify functionality, and ensure that all modules work efficiently and reliably. In the Scholarship Management System, testing helps confirm that features such as application submission, eligibility checking, approval processing, and reporting operate without issues and provide accurate results. It also ensures that data is processed correctly across different modules. Proper testing helps in identifying potential issues before deployment.

The main types of testing performed are:

- Unit Testing
- Integration Testing
- Validation Testing

These testing approaches collectively ensure that each component of the system works correctly both independently and when integrated with other modules. Through systematic testing, errors and inconsistencies are identified and resolved at an early stage, reducing the chances of system failure after deployment. It also improves system performance, enhances security, and ensures a smooth user experience for both students and administrators. Additionally, testing helps maintain data integrity and consistency throughout the system. It ensures that the system meets user expectations and functional requirements. Regular testing also supports system stability during updates and enhancements. It helps in building confidence in the system’s reliability. Overall, it contributes to the development of a robust and efficient application.

5.1.1 Unit Testing

Unit testing is performed to verify the functionality of individual modules of the Scholarship Management System independently. Each module, such as scholarship listing, application submission, eligibility evaluation, and approval, is tested separately to ensure that it performs the intended operations correctly. During this phase, inputs are given to the module and outputs are compared with expected results to detect any errors or mismatches. This process helps ensure that each unit behaves as designed and meets the required specifications.

Unit testing helps identify errors at an early stage of development and reduces the chances of issues during later stages. It improves code quality and makes debugging easier for developers. By testing each component individually, the system becomes more reliable and stable. It also ensures that modules function correctly before integration with other parts of the system.

Table 5.1: Unit Testing Test Cases

Test Case ID	Test Object	Test Data	Expected Result	Actual Result	Test Result
TC_U01	Scholarship List Module	View available scholarships	System displays all active scholarships	Scholarships displayed correctly	Pass
TC_U02	Scholarship Details Module	Click "View Details"	System shows scholarship eligibility criteria	Details displayed successfully	Pass
TC_U03	Scholarship Application Module	Student submits application form	Application record is created in the system	Application stored successfully	Pass
TC_U04	Eligibility Evaluation Module	Income, marks, attendance entered	System checks eligibility conditions	Eligibility calculated correctly	Pass
TC_U05	Student Record Module	Application approved	Student record created or updated	Student record created successfully	Pass
TC_U06	Scholarship Award Module	Application status = Approved	Scholarship award record generated	Award record created successfully	Pass

5.1.2 Integration Testing

Integration testing is conducted to verify the interaction between different modules of the Scholarship Management System. After individual modules are tested, they are combined to check whether data flows correctly between them. For example, the application module is integrated with eligibility evaluation and approval modules to ensure proper processing. This testing confirms that modules communicate effectively and function together without errors.

It also helps identify interface issues and data inconsistencies between interconnected modules. Integration testing improves overall system reliability and ensures smooth operation when multiple components interact. It plays an important role in delivering a stable and well-coordinated system.

Table 5.2: Integration Testing Test Cases

Test Case ID	Test Object	Test Data	Expected Result	Actual Result	Test Result
TC_I01	Scholarship List → Scholarship Details	Click scholarship scheme	Details page opens correctly	Page displayed successfully	Pass
TC_I02	Application Form → Database	Submit application form	Data stored in Scholarship Application object	Data stored successfully	Pass
TC_I03	Application → Eligibility Check	Submit student eligibility details	System verifies eligibility conditions	Eligibility checked correctly	Pass
TC_I04	Application → Admin Review	Admin reviews application	Application visible to admin	Admin can review successfully	Pass
TC_I05	Approval → Student Record	Admin approves application	Student record created or updated	Record updated successfully	Pass
TC_I06	Approval → Scholarship Award	Application approved	Scholarship award generated	Award record created	Pass

5.1.3 Validation Testing

Validation testing is performed to ensure that the system meets the specified functional requirements and handles user inputs correctly. This testing verifies whether the system validates the data entered by users and prevents invalid or incomplete information from being processed. In the Scholarship Management System, validation testing checks conditions such as required fields in the application form, eligibility criteria like income and academic marks, and prevention of duplicate applications. This ensures that the system processes only valid data and maintains data integrity throughout the scholarship management process. It also improves the accuracy and reliability of the system outputs by minimizing errors.

Table 5.3: Validation Testing Test Cases

Test Case ID	Test Object	Test Data	Expected Result	Actual Result	Test Result
TC_V01	Scholarship Application Form	Empty required fields	System shows validation error	Error message displayed	Pass
TC_V02	Income Field	Income above eligibility limit	System marks application as not eligible	Eligibility validation works	Pass
TC_V03	Marks Field	Marks below required percentage	Application rejected	Validation performed correctly	Pass
TC_V04	Attendance Field	Attendance less than required	Application marked ineligible	System validates attendance	Pass
TC_V05	Duplicate Application	Same student applies twice	System prevents duplicate entry	Duplicate prevented	Pass

SUMMARY

This chapter discussed the testing process carried out for the Scholarship Management System. Various testing methods such as unit testing, integration testing, and validation testing were performed to ensure that the system functions correctly and meets the specified requirements. The test results confirm that the system modules work properly, data flows correctly between components, and input validations are successfully implemented. These testing activities ensure the reliability, accuracy, and overall performance of the system.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMANTS

6.1 CONCLUSION

The Scholarship Management System provides an efficient and automated solution for managing scholarship schemes, student applications, and scholarship awards within the institution. The system simplifies the scholarship process by allowing students to view available scholarships, check eligibility criteria, and submit applications through an online portal. Administrative users can review applications, verify eligibility conditions, and approve or reject scholarships through the Salesforce platform. Automation implemented using Apex triggers ensures that student records and scholarship award records are generated automatically after approval. The system improves transparency, reduces manual workload, and ensures accurate data management. Overall, the Scholarship Management System successfully streamlines the scholarship management process and provides a reliable platform for handling scholarship-related activities.

6.2 FUTURE ENHANCEMENTS

Although the current system provides an efficient solution for scholarship management, several enhancements can be implemented in the future to improve its functionality. Future improvements may include the integration of email and SMS notifications to inform students about application status updates. Advanced reporting and dashboard features can also be added to provide administrators with detailed insights into scholarship distribution and student eligibility statistics. The system can be enhanced with AI-based recommendation features to suggest suitable scholarships for students based on their academic and financial profiles. Additionally, integration with external government scholarship portals can further expand the system's capabilities and improve the overall scholarship management process.

APPENDICES

A. SAMPLE CODING

Backend File

ScholarshipController.cls

```
public with sharing class ScholarshipController {
    @AuraEnabled(cacheable=true)
    public static List<Scholarship_Scheme__c> getActiveScholarships() {
        return [
            SELECT Id,
                   Name,
                   Course_Eligible__c,
                   Admission_Type__c,
                   Eligible_Study_Year__c,
                   Maximum_Family_Income__c,
                   Gender_Eligible__c,
                   Community_Eligible__c,
                   Description__c
            FROM Scholarship_Scheme__c
            WHERE Status__c = 'Active'
            ORDER BY Name
        ];
    }
    @AuraEnabled(cacheable=true)
    public static Scholarship_Scheme__c getScholarshipDetails(Id schemeId) {
```

```
    if(schemeId == null){
        throw new AuraHandledException('Scholarship Id is required');
    }
    Scholarship_Scheme__c scheme = [
        SELECT Id,
               Name,
               Course_Eligible__c,
               Admission_Type__c,
               Eligible_Study_Year__c,
               Maximum_Family_Income__c,
               Minimum_Marks_Required__c,
               Minimum_Attendance_Required__c,
               Gender_Eligible__c,
               Community_Eligible__c,
               Description__c
        FROM Scholarship_Scheme__c
        WHERE Id = :schemeId
        LIMIT 1
    ];
    return scheme;
  }
  @AuraEnabled(cacheable=true)
public static List<Scholarship_Scheme__c> recommendScholarships(
  Decimal income,
```

```
    Decimal marks,
    Decimal attendance,
    String gender,
    String community,
    String year
){
    List<Scholarship_Scheme__c> schemes = [
        SELECT Id,
               Name,
               Description__c,
               Maximum_Family_Income__c,
               Minimum_Marks_Required__c,
               Minimum_Attendance_Required__c,
               Gender_Eligible__c,
               Community_Eligible__c,
               Eligible_Study_Year__c
        FROM Scholarship_Scheme__c
        WHERE Status__c = 'Active'
    ];
    List<Scholarship_Scheme__c> recommended = new List<Scholarship_Scheme__c>();
    for(Scholarship_Scheme__c s : schemes){
        Boolean eligible = true;
        if(s.Maximum_Family_Income__c != null && income >
s.Maximum_Family_Income__c)
```

```
            eligible = false;
        if(s.Minimum_Marks_Required__c != null && marks <
s.Minimum_Marks_Required__c)
            eligible = false;
        if(s.Minimum_Attendance_Required__c != null && attendance <
s.Minimum_Attendance_Required__c)
            eligible = false;
        if(s.Gender_Eligible__c != 'Both' && s.Gender_Eligible__c != gender)
            eligible = false;
        if(s.Community_Eligible__c != null &&
!s.Community_Eligible__c.contains(community))
            eligible = false;
        if(s.Eligible_Study_Year__c != null && !s.Eligible_Study_Year__c.contains(year))
            eligible = false;
        if(eligible){
            recommended.add(s);
        }
    }
    return recommended;
}
}
```

Frontend File

scholarshipApplicationForm.html

```
<template>
    <lightning-card title="Scholarship Application">
```

```
<div class="slds-p-horizontal_small slds-p-top_small">
    <lightning-button
        variant="neutral"
        label="Back"
        onclick={handleBack}
        icon-name="utility:back">
    </lightning-button>
</div>

<lightning-record-edit-form
    object-api-name="Scholarship_Application__c"
    onsuccess={handleSuccess}
    onsubmit={handleSubmit}>
    <lightning-messages></lightning-messages>
    <lightning-input-field field-name="Scholarship_Scheme__c"
                        value={schemeId}>
    </lightning-input-field>
    <lightning-input-field field-name="Student_Name__c"></lightning-input-field>
    <lightning-input-field field-name="Roll_Number__c"></lightning-input-field>
    <lightning-input-field field-name="Date_of_Birth__c"></lightning-input-field>
    <lightning-input-field field-name="Gender__c"
                        onchange={handleFieldChange}>
    </lightning-input-field>
    <lightning-input-field field-name="Religion__c"></lightning-input-field>
    <lightning-input-field field-name="Community_Caste__c"
                        onchange={handleFieldChange}>
    </lightning-input-field>
    <lightning-input-field field-name="Course__c"></lightning-input-field>
```

```
<lightning-input-field field-name="Academic_Year__c"></lightning-input-field>
<lightning-input-field field-name="Current_Year__c"
            onchange={handleFieldChange}>
</lightning-input-field>
<lightning-input-field field-name="Institution__c"></lightning-input-field>
<lightning-input-field field-name="Date_of_Joining__c"></lightning-input-field>

<lightning-input-field field-name="Previous_Qualification__c"></lightning-input-field>
<lightning-input-field field-name="Previous_Year_Percentage__c"
            onchange={handleFieldChange}>
</lightning-input-field>
<lightning-input-field field-name="Attendance_Percentage__c"
            onchange={handleFieldChange}>
</lightning-input-field>
<lightning-input-field field-name="Counselling_Number__c"></lightning-input-field>
<lightning-input-field field-name="X10th_Register_Number__c"></lightning-input-field>
<lightning-input-field field-name="X12th_Register_Number__c"></lightning-input-field>
<lightning-input-field field-name="Aadhaar_Number__c"></lightning-input-field>
<lightning-input-field field-name="Annual_Family_Income__c"
            onchange={handleFieldChange}>
</lightning-input-field>
<lightning-input-field field-name="First_Graduate_in_Family__c"></lightning-input-field>
<lightning-input-field field-name="Father_Name__c"></lightning-input-field>
<lightning-input-field field-name="Father_Occupation__c"></lightning-input-field>
```

```
<lightning-input-field field-name="Father_Mobile__c"></lightning-input-field>
<lightning-input-field field-name="Mother_Name__c"></lightning-input-field>
<lightning-input-field field-name="Mother_Occupation__c"></lightning-input-field>
<lightning-input-field field-name="Mother_Mobile__c"></lightning-input-field>
<lightning-input-field field-name="Permanent_Address__c"></lightning-input-field>
<lightning-input-field field-name="Current_Address__c"></lightning-input-field>
<lightning-input-field field-name="Distance_to_College__c"></lightning-input-field>
<lightning-input-field field-name="Student_Mobile__c"></lightning-input-field>
<lightning-input-field field-name="Email_ID__c"></lightning-input-field>
<lightning-input-field field-name="Bank_Name__c"></lightning-input-field>
<lightning-input-field field-name="Branch__c"></lightning-input-field>
<lightning-input-field field-name="Account_Number__c"></lightning-input-field>
<lightning-input-field field-name="IFSC_Code__c"></lightning-input-field>
<lightning-input-field field-name="MICR_Code__c"></lightning-input-field>
<lightning-input-field field-name="Hosteller__c"></lightning-input-field>
<lightning-input-field field-name="Disability_Percentage__c"></lightning-input-field>
<lightning-input-field field-name="CAPF_Ward__c"></lightning-input-field>
<lightning-input-field field-name="Orphan__c"></lightning-input-field>
<lightning-input-field field-name="COVID_Parent_Loss__c"></lightning-input-field>
<div class="slds-m-top_medium">
   <lightning-button
      type="submit"
      variant="brand"
      label="Submit Application">
   </lightning-button>
```

```
      </div>
    </lightning-record-edit-form>
    <template if:true={isEligible}>
      <div class="slds-text-color_success slds-p-around_small">
        Eligible for this scholarship
      </div>
    </template>
    <template if:false={isEligible}>
      <div class="slds-text-color_error slds-p-around_small">
        {errorMessage}
      </div>
    </template>
  </lightning-card>
</template>
```

scholarshipApplicationForm.js

```
import { LightningElement, api, track } from 'lwc';
import { ShowToastEvent } from 'lightning/platformShowToastEvent';
import checkEligibility from
'@salesforce/apex/ScholarshipEligibilityEngine.checkEligibility';
export default class ScholarshipApplicationForm extends LightningElement {
  @api schemeId;
  @track isEligible = false;
  @track errorMessage = '';
  income;
  marks;
  attendance;
  gender;
  community;
  year;
```

```
handleFieldChange(event) {
    const field = event.target.fieldName;
    const value = event.detail.value;
    if(field === 'Annual_Family_Income__c'){
        this.income = value;
    }
    if(field === 'Previous_Year_Percentage__c'){
        this.marks = value;
    }
    if(field === 'Attendance_Percentage__c'){
        this.attendance = value;
    }
    if(field === 'Gender__c'){
        this.gender = value;
    }
    if(field === 'Community_Caste__c'){
        this.community = value;
    }
    if(field === 'Current_Year__c'){
        this.year = value;
    }
    this.checkEligibilityHandler();
checkEligibilityHandler(){
    if(!this.schemeId) return;
    checkEligibility({
        schemeId: this.schemeId,
        income: this.income,
        marks: this.marks,
```

```
            attendance: this.attendance,
            gender: this.gender,
            community: this.community,
            year: this.year
        })
        .then(result => {
            if(result === 'Eligible'){
                this.isEligible = true;
                this.errorMessage = '';
            }
            else{
                this.isEligible = false;
                this.errorMessage = result;
            }
        })
        .catch(error=>{
            console.error(error);
            this.isEligible = false;
            this.errorMessage = 'An error occurred while checking eligibility. Please try again.';
        });
    }
    handleSubmit(event) {
        const fields = event.detail.fields;
        if (this.schemeId) {
            fields.Scholarship_Scheme__c = this.schemeId;
        }
        if (!this.isEligible && this.schemeId) {
            event.preventDefault();
```

```
            this.dispatchEvent(
                new ShowToastEvent({
                    title: 'Not Eligible',
                    message: 'You are not eligible for this scholarship',
                    variant: 'error'
                })
            );
        }
    }
    handleBack() {
        const evt = new CustomEvent('back', {
            bubbles: true,
            composed: true
        });
        this.dispatchEvent(evt);
    }
}
```

scholarshipApplicationForm.js-meta.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
    <apiVersion>65.0</apiVersion>
    <isExposed>true</isExposed>
    <targets>
        <target>lightningCommunity__Page</target>
        <target>lightning__AppPage</target>
    </targets>
</LightningComponentBundle>
```

B. SCREENSHOTS

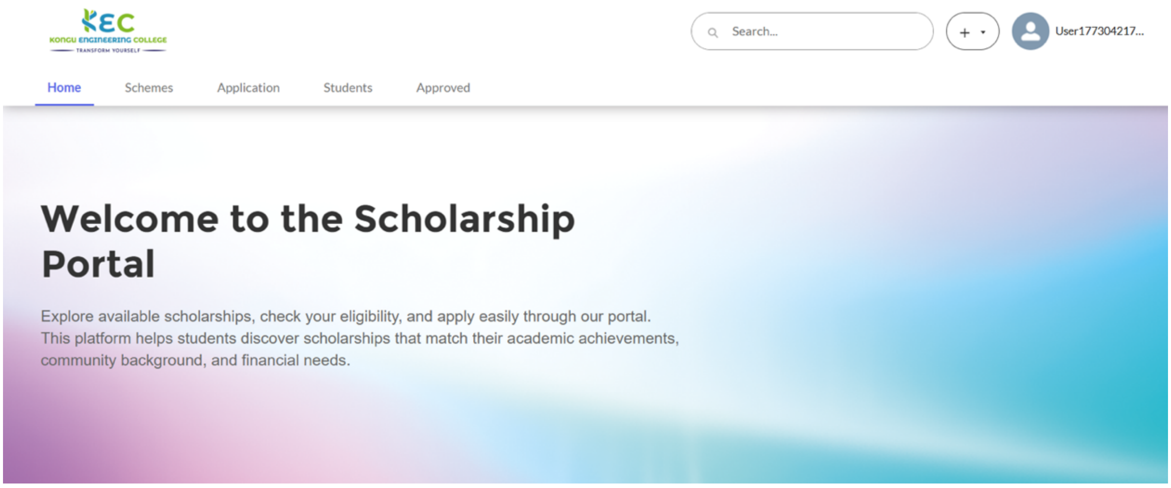

Figure B.1 Scholarship Portal Home Page

The main portal page where students access the scholarship system, browse available schemes, and navigate features like application submission, status tracking, and profile management through a user-friendly interface.

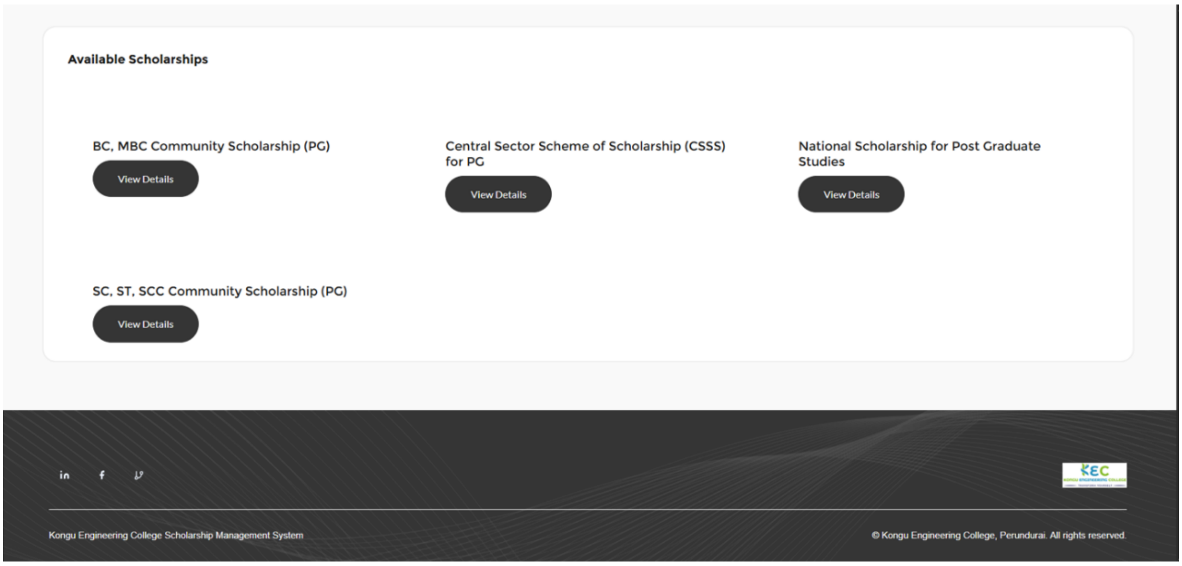

Figure B.2 Scholarship List Page

Displays all available scholarship schemes with essential details, including scholarship name, eligibility criteria, and related information. It helps students easily explore and identify suitable scholarships based on their qualifications.

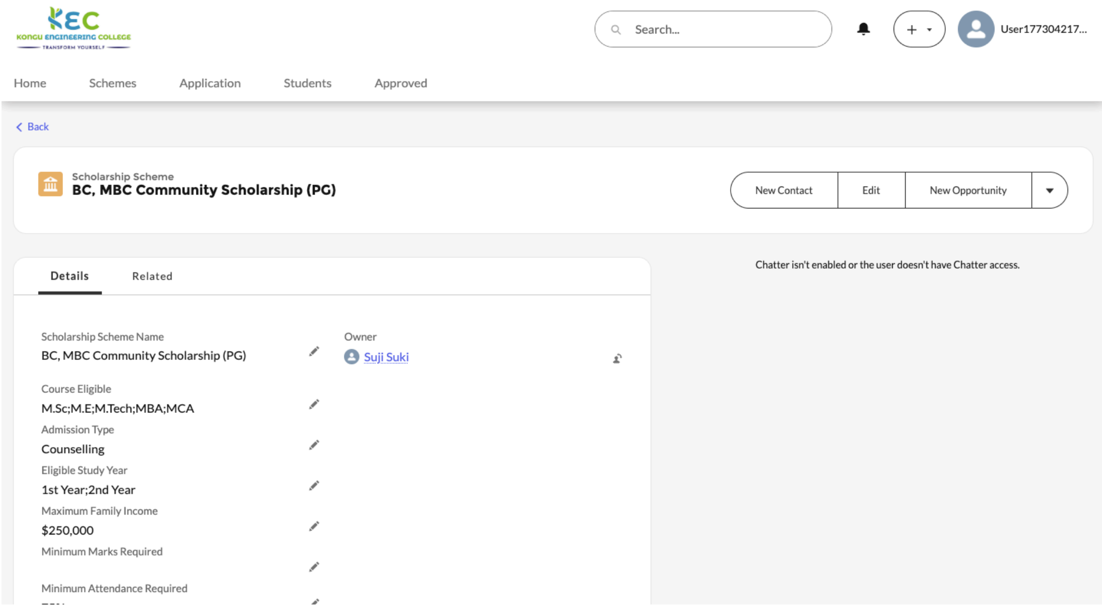

Figure B.3 Scholarship Details Page

Shows detailed information about a selected scholarship scheme including eligibility requirements and description, and benefits. It helps students understand criteria clearly and make informed decisions before applying for scholarships.

Figure B.4 Scholarship Eligibility Check Page

Screen where students enter details such as income, marks, attendance, and community to check eligibility for scholarships. It evaluates inputs and helps students identify suitable schemes based on their qualifications.

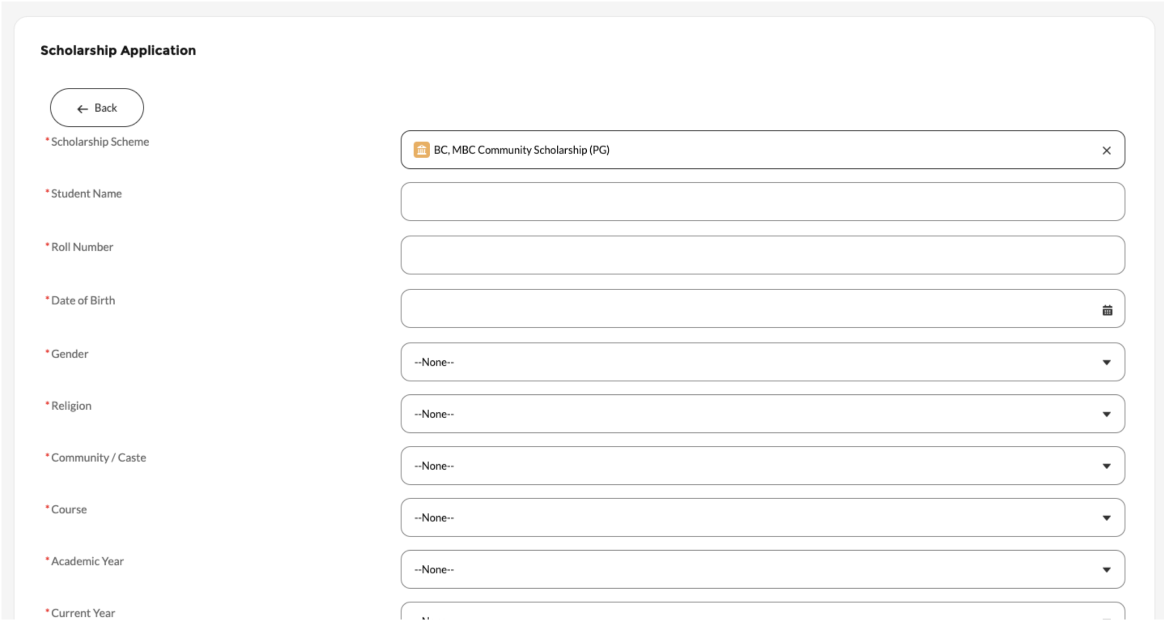

Figure B.5 Scholarship Application Form

Scholarship application form where students enter personal, academic, and financial details to apply for a scholarship. It ensures accurate data submission and supports efficient processing and evaluation of applications.

Figure B.6 Scholarship Officer Login Page

Login interface used by the scholarship officer to access the administration panel. It requires valid credentials and ensures authorized access to manage applications, scholarships, and system operations.

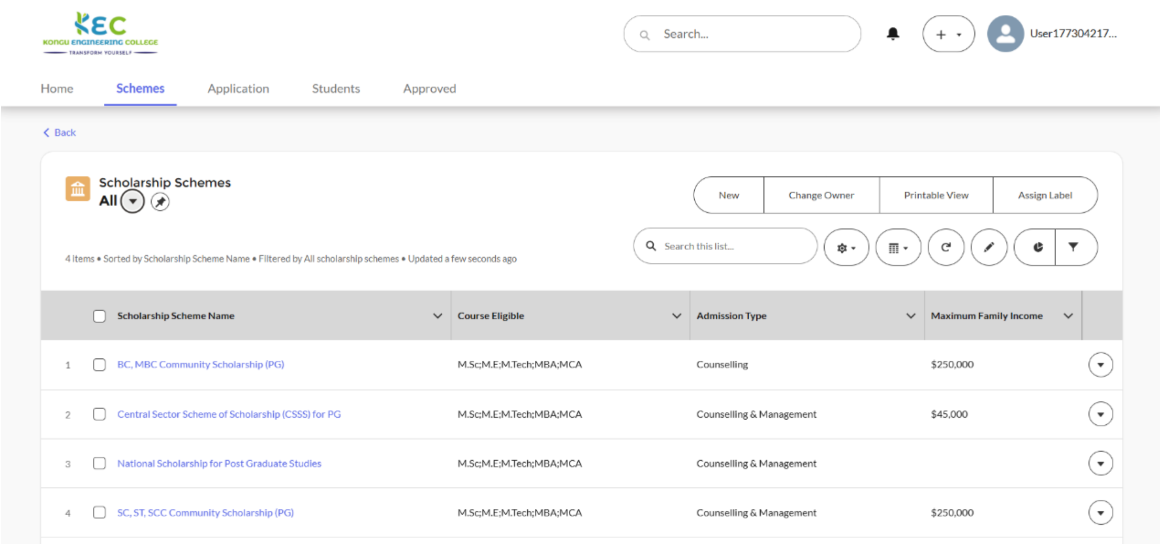

Figure B.7 Scholarship Scheme Management Page

Screen used by the admin or officer to create, update, or manage scholarship schemes. It allows modification of eligibility criteria, scholarship amount, and application deadlines. This feature ensures efficient management and maintenance of scholarship programs.

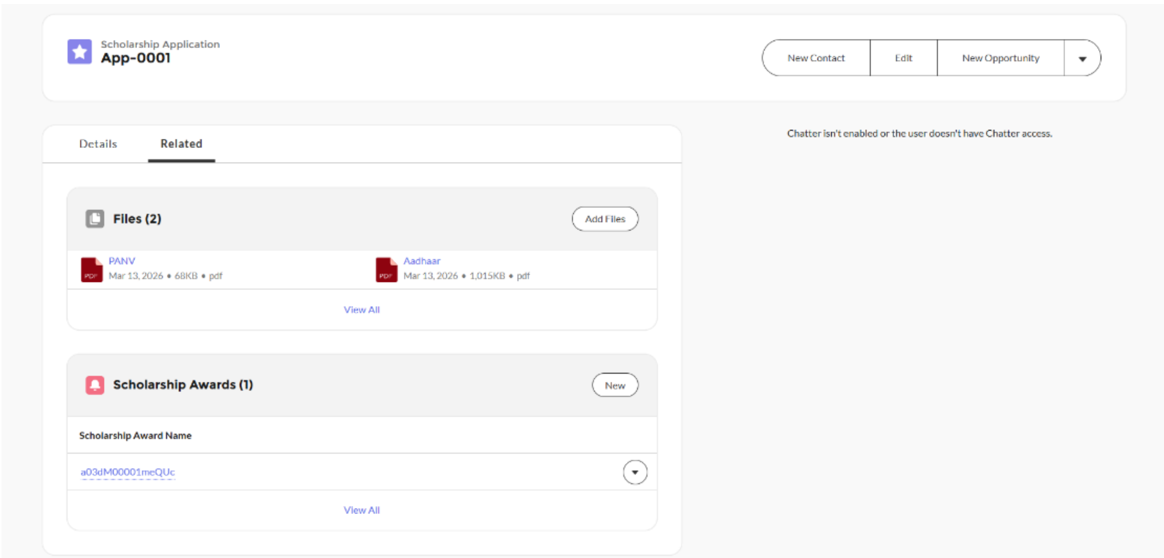

Figure B.8 Student Application Review Page

Displays student applications for review and verification by the scholarship officer, including applicant details, submitted documents, and eligibility information. This feature enables efficient screening and supports accurate decision-making during the evaluation process.

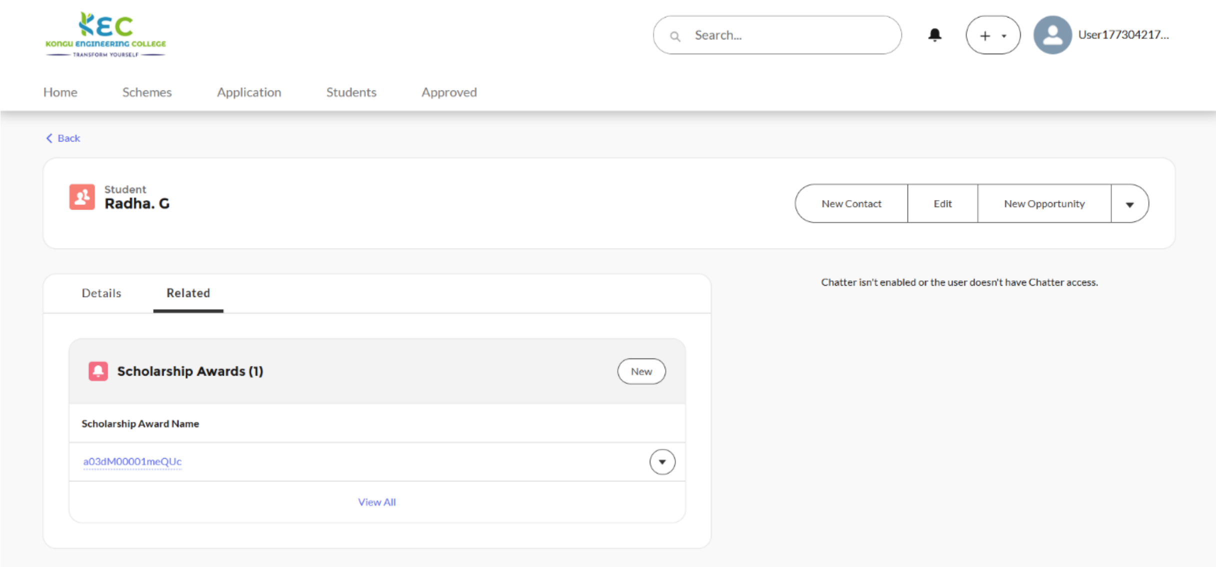

Figure B.9 Student details page

Displays details of students whose scholarship applications have been approved. It includes information such as student name, eligibility criteria, and scholarship details. This feature helps administrators efficiently verify and manage approved candidates.

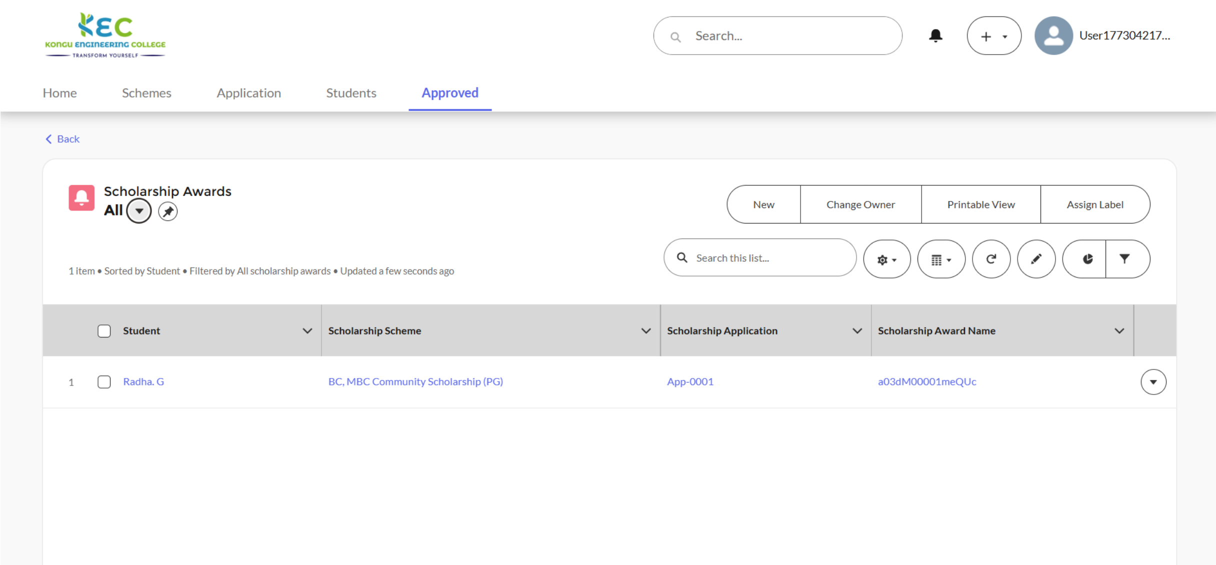

Figure B.10 Scholarship Award Record Page

Displays the scholarship awards generated after the approval of student applications. It provides details such as awarded amount, student information, and scholarship type. This feature ensures transparency and enables students to view their awarded benefits clearly.

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SUSTAINABLE DEVELOPMENT GOALS (SDGS)

SDG 4: QUALITY EDUCATION

Target 4.3: Equal access to affordable and quality higher education

- The system provides a centralized platform where students can easily discover and apply for scholarships.
- It ensures that financial constraints do not restrict students from pursuing higher education opportunities.

Target 4.5: Eliminate disparities in education access

- The platform supports inclusive access for students from SC/ST, minority, and economically weaker sections.
- It promotes equal opportunities by simplifying and standardizing the scholarship application process.

SDG 10: REDUCED INEQUALITIES

Target 10.2: Empower and promote inclusion of all individuals

- The system enables equal access to scholarship information regardless of socio-economic background.
- It ensures that students from rural and underprivileged communities are included in educational opportunities.

Target 10.3: Ensure equal opportunity and reduce inequalities of outcome

- The platform provides a transparent and unbiased scholarship allocation process.
- It minimizes manual errors and discrimination through a structured and digital evaluation system.

DATE : **SUPERVISOR SIGNATURE**